

MODULE 4:

Azure Active Directory .

Introduction to Active Directory (AD), Identity and Authentication in Public Cloud – Introduction to Azure AD – Extending Active Directory into Azure – Azure AD and applications – Reporting and Monitoring Azure AD – Multi Factor Authentication – Messaging –Biz Talk Services, Media Services . Azure Cost Management and Optimization- Azure Cost Management Tools - Budgeting and Forecasting in Azure - Cost Optimization Strategies

Day 1: Azure Active Directory Fundamentals (60 minutes)

Task 1: Understanding Active Directory Concepts

Research and Document:

1. Traditional Active Directory vs Azure AD:

- Create a comparison table covering:
 - Architecture (on-premises vs cloud)
 - Authentication protocols (Kerberos/NTLM vs SAML/OAuth)
 - Primary use cases
 - Management approach
 - Integration capabilities

2. Identity and Authentication in Public Cloud:

- Define: Identity, Authentication, Authorization
- Explain: Federation, SSO (Single Sign-On), MFA
- Cloud identity models: Cloud-only, Hybrid, Federated

3. Azure AD Editions:

Feature	Free	Office 365	Premium P1	Premium P2
Users	500,000	Unlimited	Unlimited	Unlimited
SSO Apps	10	Unlimited	Unlimited	Unlimited
MFA	☐	✓	✓	✓

Conditional Access	☐	☐	✓	✓
Identity Protection	☐	☐	☐	✓
Price per user/month	Free	Included	~\$6	~\$9

Complete this table and add 5 more feature comparisons.

Task 2: Create and Configure Azure AD Tenant

1. Access Azure AD:

- Navigate to Azure Active Directory in Azure Portal
- Document your tenant details:
 - Tenant name
 - Tenant ID
 - Primary domain
 - Default directory

Custom Domain Setup:

- Add a custom domain (if you have one, or document the process)
- Steps to verify domain ownership
- DNS record requirements
- Screenshot the domain verification process

Tenant Branding:

- Configure company branding:
 - Add company logo
 - Background image
 - Sign-in page text
 - Username hint
- Screenshot the customized sign-in page

Questions:

- What is a tenant in Azure AD?
- Can you have multiple Azure AD tenants?
- What's the difference between a tenant and a subscription?

Task 3: User and Group Management

Create the following organizational structure:

Users (Create 15 users):

User	Department	Job Title	Location
john.admin@[yourdomain]	IT	IT Administrator	New York
sarah.hr@[yourdomain]	HR	HR Manager	New York
mike.finance@[yourdomain]	Finance	Finance Manager	London
lisa.sales@[yourdomain]	Sales	Sales Director	Tokyo
tom.dev@[yourdomain]	IT	Developer	New York
emma.dev@[yourdomain]	IT	Developer	London
alex.marketing@[yourdomain]	Marketing	Marketing Manager	New York
sophie.support@[yourdomain]	IT	Support Specialist	Tokyo

[Create 7 more users across departments]

For each user, configure:

- First name, Last name
- User principal name
- Job title and Department
- Usage location
- Office location
- Manager (hierarchical structure)
- Profile photo (optional)

Groups (Create the following):**Security Groups:**

1. IT-Administrators
2. HR-Team
3. Finance-Team
4. Sales-Team
5. Developers
6. Marketing-Team
7. Support-Team

Office 365 Groups:

1. All-Employees
2. Project-Phoenix (dynamic membership)
3. Executive-Team

Dynamic Groups (Create at least 2):

Example dynamic membership rule:

`(user.department -eq "IT") -and (user.jobTitle -contains "Developer")`

Create dynamic groups for:

- All users in New York
- All managers (job title contains "Manager")

Task 4: Administrative Units

1. Create Administrative Units:
 - AU-NewYork (users from New York)
 - AU-London (users from London)
 - AU-Tokyo (users from Tokyo)
2. Assign users to appropriate AUs
3. Delegate administration:
 - Assign "User Administrator" role scoped to AU-NewYork to john.admin

Questions:

- What are Administrative Units used for?
- How do they differ from Groups?
- What are the benefits for large organizations?

Day 2: Azure AD Roles and RBAC (50 minutes)

Task 1: Understanding Azure AD Roles

Create a comprehensive table:

Role Name	Permissions	Use Case	Assign To
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Global Administrator	Full access	Emergency only	john.admin
User Administrator	Manage users/groups	HR operations	sarah.hr
Application Administrator	Manage apps	IT team	[User]
Security Administrator	Security settings	Security team	[User]
Helpdesk Administrator	Reset passwords	Support team	sophie.support

Research and add 10 more common Azure AD roles.

Task 2: Role Assignments

- 1. Assign built-in roles:**
 - **Global Administrator** → john.admin (for lab purposes)
 - **User Administrator** → sarah.hr
 - **Helpdesk Administrator** → sophie.support
 - **Security Reader** → mike.finance
 - **Application Administrator** → tom.dev
- 2. Test role permissions:**
 - **Sign in as sarah.hr**
 - **Try to create a new user (should work)**
 - **Try to modify subscription settings (should fail)**
 - **Document results**
- 3. Sign in as sophie.support:**
 - **Reset password for a test user**
 - **Try to delete a user (should fail)**
 - **Document results**

Screenshot Required:

- Role assignments page
- Successful and failed actions from different user contexts

Task 3: Privileged Identity Management (PIM) - Documentation

Research and document:

1. What is PIM?
 - Purpose and benefits
 - Just-in-time privileged access
 - Approval workflows
2. PIM Setup Process:
 - Required license (Premium P2)
 - Configuration steps
 - Activation workflow
3. Best Practices:
 - Time-bound role assignments
 - Approval requirements
4. Access reviews

Create a PIM implementation plan for your organization (500 words).

Day 3: Extending Active Directory to Azure (70 minutes)

Task 1: Hybrid Identity Scenarios

Research and document three hybrid identity scenarios:

1. Password Hash Synchronization (PHS)
 - How it works
 - Authentication flow diagram
 - Pros and cons
 - Use cases
2. Pass-through Authentication (PTA)
 - Architecture
 - Authentication flow
 - Comparison with PHS
 - When to use
3. Federation (AD FS)

- Federation concept
- AD FS architecture
- Claims-based authentication
- Use cases

Create comparison matrix:

Feature	PHS	PTA	Federation
On-prem infrastructure required	Minimal	Agent only	Full AD FS
Password in cloud	Hash only	No	No
Complexity	Low	Medium	High
Seamless SSO	Yes	Yes	Yes
MFA support	Azure MFA	Azure MFA	Both
Best for	Most scenarios	High security	Complex reqs

Task 2: Azure AD Connect (Simulation/Documentation)

Note: If you don't have an on-premises AD, document the process thoroughly.

Document the Azure AD Connect setup process:

- 1. Pre-requisites:**
 - Windows Server requirements
 - AD permissions needed
 - Network requirements (ports, URLs)
 - Account requirements
- 2. Installation Options:**
 - Express settings vs Custom
 - Synchronization options
 - Sign-in options
 - Filtering options
- 3. Create a deployment plan:**
 - Server specifications
 - Network diagram showing on-prem AD and Azure AD
 - Synchronization scope (which OUs to sync)
- 4. Attribute mapping**
- 5. Filtering strategy**

Post-installation tasks:

- Verify synchronization
- Configure sync scheduler
- Password writeback
- Device writeback

If you have access to a test environment:

- 1. Install Azure AD Connect on a Windows Server**
- 2. Configure with Express settings**
- 3. Sync users from on-premises AD to Azure AD**
- 4. Verify synchronized users appear in Azure AD**
- 5. Test sign-in with synchronized account**
- 6. Document the entire process with screenshots**

Questions:

- What is the sync cycle interval?
- Can you selectively sync users?
- What happens when you delete a user on-premises?
- How does password sync work?

Task 3: Seamless Single Sign-On (SSO)

Research and document:

1. How Seamless SSO works
2. Technical requirements
3. Configuration steps
4. Supported scenarios
5. Limitations

Create a user experience flow diagram showing:

- User on domain-joined machine
- Accessing cloud application
- Authentication flow with Seamless SSO
- Kerberos ticket exchange

Day 4: Azure AD Application Integration (80 minutes)

Task 1: Enterprise Application Gallery

1. Add applications from gallery:

Application 1: Microsoft 365

- Add Microsoft 365 (if not already present)
- Assign users: All-Employees group
- Configure user settings

Application 2: Salesforce (Trial)

- Add Salesforce from gallery
- Configure SSO (SAML-based)
- User provisioning setup
- Assign users: Sales-Team group

Application 3: Dropbox Business

- Add Dropbox Business
- Configure SAML SSO
- Test sign-in

Application 4: Slack

- Add Slack
 - Configure SSO
 - Automatic user provisioning
2. Document for each application:
 - SSO configuration type (SAML, OAuth, etc.)

- User assignment method
- Provisioning configuration
- Test results

Screenshot Required:

- Enterprise applications list
- SSO configuration page
- User assignments

Task 2: Custom Application Registration

Register a custom application:

1. **App Registration 1: Web Application**
 - Name: CompanyPortal-WebApp
 - Supported account types: Single tenant
 - Redirect URI: <https://localhost:44321/signin-oidc>
 - Generate client secret
 - Configure API permissions:
 - Microsoft Graph: User.Read
 - Microsoft Graph: Directory.Read.All
 - Grant admin consent
2. **App Registration 2: Single Page Application (SPA)**
 - Name: CompanyPortal-SPA
 - Platform: Single-page application
 - Redirect URI: <http://localhost:3000>
 - Implicit flow: ID tokens, Access tokens
3. **App Registration 3: Mobile/Desktop App**
 - Name: CompanyMobile-App
 - Platform: Mobile and desktop
 - Redirect URI: Custom scheme (e.g., myapp://auth)

For each app registration, document:

- Application (client) ID
- Directory (tenant) ID
- Client secrets/certificates
- API permissions granted
- Authentication flows enabled